

## WARFIELD, BC, Canada

WMO: 714010

Lat: 49.1123N

Lon: 117.7386W

Elev: 567

StdP: 94.70

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | -13.5      | -10.8 | -18.2                         | 0.8 | -12.1 | -15.5 | 1.0 | -9.3 | 6.2                   | -3.7 | 5.5 | -3.0 | 1.8                   | 320  | 0.521 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.6                   | 34.5            | 17.2 | 32.5 | 16.7 | 30.4 | 16.1 | 18.6                | 29.9 | 17.6 | 28.9 | 16.8 | 27.8 | 2.1                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 15.0                            | 11.4 | 20.0 | 14.1 | 10.7 | 19.8 | 13.1 | 10.1 | 18.9 | 54.7          | 30.1 | 51.7 | 28.9 | 49.1 | 28.0 | 22.6           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 5.5           | 4.8  | 4.2 |                            | -16.6 | 37.9               | 3.1 | 1.7                                                | -18.9 | 39.1       | -20.7 | 40.1       | -22.4 | 41.0       | -24.7 | 42.2 |     |
| (5)               |      |     | DB                         | -16.6 | 20.2               | 3.2 | 1.4                                                | -18.9 | 21.2       | -20.8 | 22.0       | -22.5 | 22.8       | -24.8 | 23.8 |     |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.9   | -2.1  | -0.5  | 3.9   | 8.5   | 13.7  | 16.9  | 21.4  | 21.0  | 15.5  | 7.8   | 1.8   | -2.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 9.17  | 4.14  | 3.72  | 3.44  | 3.38  | 3.92  | 3.61  | 3.45  | 3.03  | 3.67  | 3.59  | 3.75  | 3.82 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1657  | 376   | 295   | 191   | 67    | 11    | 1     | 0     | 0     | 3     | 85    | 246   | 383  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3714  | 634   | 528   | 448   | 294   | 153   | 69    | 13    | 11    | 101   | 326   | 496   | 642  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1238  | 0     | 0     | 1     | 24    | 124   | 207   | 354   | 342   | 167   | 18    | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 253   | 0     | 0     | 0     | 0     | 8     | 26    | 109   | 94    | 16    | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 3394  | 0     | 0     | 0     | 11    | 162   | 394   | 1423  | 1182  | 222   | 1     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1509  | 0     | 0     | 0     | 1     | 46    | 143   | 706   | 543   | 72    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.8   | 1.7   | 1.8   | 1.9   | 2.1   | 1.9   | 1.8   | 1.7   | 1.7   | 1.6   | 1.6   | 1.6   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 697   | 85    | 48    | 52    | 47    | 65    | 66    | 34    | 28    | 36    | 56    | 76    | 92   | (15) |
| (16) |                                                                                  | PrecMax     | 906   | 144   | 101   | 117   | 85    | 142   | 111   | 83    | 86    | 132   | 93    | 144   | 153  | (16) |
| (17) |                                                                                  | PrecMin     | 554   | 28    | 12    | 19    | 18    | 20    | 34    | 0     | 5     | 4     | 5     | 26    | 41   | (17) |
| (18) |                                                                                  | PrecStd     | 107   | 23    | 21    | 25    | 19    | 35    | 18    | 25    | 23    | 28    | 22    | 30    | 28   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 5.6   | 9.0   | 17.0  | 25.0  | 30.6  | 34.3  | 37.4  | 36.4  | 32.0  | 21.2  | 11.6  | 6.3  | (19) |
| (20) |                                                                                  |             | MCWB  | 3.5   | 4.7   | 8.4   | 12.5  | 14.9  | 17.9  | 17.8  | 17.3  | 15.7  | 12.4  | 9.2   | 3.9  | (20) |
| (21) |                                                                                  | 2%          | DB    | 3.6   | 6.4   | 13.7  | 21.1  | 27.7  | 30.6  | 35.2  | 34.0  | 28.8  | 17.8  | 9.4   | 4.1  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.2   | 3.4   | 6.7   | 10.2  | 14.1  | 16.5  | 17.4  | 16.9  | 15.5  | 10.8  | 7.8   | 2.7  | (22) |
| (23) |                                                                                  | 5%          | DB    | 2.6   | 5.0   | 11.2  | 18.1  | 24.7  | 27.7  | 33.2  | 32.0  | 25.8  | 15.4  | 7.8   | 2.8  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.7   | 2.6   | 5.7   | 9.0   | 12.8  | 15.2  | 17.2  | 16.4  | 14.2  | 9.9   | 6.3   | 2.0  | (24) |
| (25) |                                                                                  | 10%         | DB    | 1.9   | 3.7   | 9.2   | 15.6  | 22.0  | 25.2  | 31.0  | 29.9  | 22.9  | 13.3  | 6.5   | 1.8  | (25) |
| (26) |                                                                                  |             | MCWB  | 1.2   | 2.0   | 4.8   | 7.8   | 11.7  | 14.3  | 16.5  | 15.8  | 13.0  | 9.2   | 5.1   | 1.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 4.1   | 5.7   | 9.7   | 13.3  | 16.3  | 19.2  | 20.3  | 19.2  | 17.7  | 13.9  | 10.1  | 4.8  | (27) |
| (28) |                                                                                  |             | MCDWB | 5.2   | 7.4   | 14.8  | 23.7  | 26.7  | 31.0  | 32.4  | 30.6  | 27.4  | 17.6  | 11.1  | 5.5  | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.7   | 4.0   | 7.9   | 11.1  | 14.8  | 17.2  | 18.9  | 18.0  | 16.1  | 12.4  | 8.2   | 3.0  | (29) |
| (30) |                                                                                  |             | MCDWB | 3.3   | 5.7   | 12.1  | 19.3  | 24.6  | 27.8  | 30.3  | 30.1  | 25.2  | 15.9  | 9.1   | 3.4  | (30) |
| (31) |                                                                                  | 5%          | WB    | 2.0   | 3.1   | 6.5   | 9.8   | 13.7  | 16.1  | 18.0  | 17.1  | 15.1  | 11.1  | 6.6   | 2.0  | (31) |
| (32) |                                                                                  |             | MCDWB | 2.4   | 4.4   | 10.3  | 16.6  | 22.3  | 25.0  | 29.1  | 29.0  | 23.5  | 14.6  | 7.5   | 2.4  | (32) |
| (33) |                                                                                  | 10%         | WB    | 1.3   | 2.2   | 5.3   | 8.5   | 12.7  | 15.2  | 17.1  | 16.4  | 14.1  | 9.7   | 5.1   | 1.2  | (33) |
| (34) |                                                                                  |             | MCDWB | 1.7   | 3.5   | 8.5   | 14.3  | 19.8  | 22.8  | 28.4  | 27.6  | 21.7  | 12.9  | 6.1   | 1.6  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 3.6   | 6.1   | 8.5   | 11.6  | 13.3  | 13.1  | 16.6  | 16.9  | 13.4  | 9.0   | 4.7   | 3.5  | (35) |
| (36) |                                                                                  |             | MCDBR | 3.7   | 7.5   | 12.4  | 16.8  | 18.3  | 18.3  | 20.4  | 20.3  | 18.1  | 12.9  | 6.4   | 4.1  | (36) |
| (37) |                                                                                  |             | MCWBR | 2.9   | 4.9   | 6.8   | 7.9   | 7.5   | 7.0   | 6.9   | 6.9   | 7.1   | 6.7   | 4.7   | 3.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 3.4   | 6.3   | 10.9  | 15.0  | 15.3  | 15.6  | 18.0  | 17.9  | 15.9  | 10.6  | 5.7   | 3.4  | (38) |
| (39) |                                                                                  |             | MCWBR | 2.7   | 4.3   | 6.5   | 7.5   | 6.6   | 6.4   | 6.6   | 6.5   | 6.6   | 5.8   | 4.5   | 3.0  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.280 | 0.280 | 0.291 | 0.339 | 0.369 | 0.361 | 0.359 | 0.370 | 0.341 | 0.317 | 0.300 | 0.273 | (40) |      |
| (41) |                                                                                  | taud        | 2.308 | 2.332 | 2.397 | 2.377 | 2.360 | 2.420 | 2.437 | 2.400 | 2.476 | 2.531 | 2.458 | 2.366 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 776   | 877   | 929   | 909   | 891   | 899   | 895   | 867   | 862   | 829   | 756   | 738   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 70    | 86    | 96    | 110   | 117   | 111   | 108   | 107   | 90    | 72    | 61    | 58    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.96  | 1.90  | 2.87  | 4.37  | 5.40  | 5.69  | 6.80  | 5.70  | 4.03  | 2.31  | 1.06  | 0.76  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.13  | 0.23  | 0.37  | 0.39  | 0.44  | 0.52  | 0.52  | 0.44  | 0.41  | 0.30  | 0.15  | 0.10  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (17 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +78     | N/A     |

Nomenclature: See separate page